

Chaima Ben Rayana

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Teboulba, Monastir, Tunisia

SUMMARY

Final-year engineering student at ENET'Com, specializing in Electronics and Communications Engineering, with a strong interest in artificial intelligence applied to biomedical signal processing and intelligent embedded systems. Hands-on experience in deep learning, full-stack development, and data analysis, backed by concrete projects in ECG/EMG processing and web applications.

TECHNICAL SKILLS

AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Biomedical Signal Processing (ECG, EMG)

Embedded & IoT: STM32, Raspberry Pi, Simulink, MATLAB, Edge AI, GPIO Programming

Web Development: ReactJS, Laravel, PHP, HTML/CSS, JavaScript, SQL, REST APIs, Full-Stack Development

DevOps & Tools: Java, Spring Boot, Spring Security (JWT), PostgreSQL, MinIO, Docker, Git, GitHub, VS Code, Figma

Languages: Arabic (native), French (C1), English (B2)

PROFESSIONAL EXPERIENCE

Software Engineering Intern – Intelligent HR Assistant

June 2026

Proxym IT

Sousse, Tunisia (Remote)

- Developed the backend API of an AI-powered HR management platform using Java / Spring Boot, Spring Security (JWT), and PostgreSQL, covering leave management, document handling, and candidate interview workflows
- Integrated the Google Gemini API to automate candidate screening, replacing manual resume review steps with AI-driven scoring logic
- Implemented secure object storage with MinIO and built RESTful endpoints supporting the end-to-end HR workflow
- Built the React/Vite frontend for AI-driven interview pages, including per-answer scoring and admin dashboards

Data Analysis Intern – HR Mobility Department

July 2025

Dräxlmaier Group

Sousse, Tunisia

- Analyzed personnel transportation data in Excel to identify logistical inefficiencies across employee shuttle routes
- Built data visualizations and analytical reports to support managerial decision-making on transportation planning and cost efficiency

ACADEMIC & PERSONAL PROJECTS

Cardiac Anomaly Detection via ECG

2025 – 2026

Deep Learning – Biomedical Signal Processing

Python, TensorFlow, NumPy, Streamlit

- Designed an arrhythmia classification system on the MIT-BIH dataset using a CNN-BiLSTM architecture with attention mechanism
- Built a complete pipeline: Pan-Tompkins segmentation, normalization, data augmentation, and per-class threshold optimization
- Results:** Accuracy 98.27% | Macro F1-score 91.08% | AUC-ROC 98.92% | Real-time Streamlit interface

Interactive Interface for EMG Feature Evaluation

2026

IoT & Signal Processing

Python, Signal Processing Libraries

- Developed an interactive interface for real-time extraction and visualization of EMG signal features.
- Implemented filtering and feature-extraction pipelines to process biomedical signals for muscular performance monitoring
- Designed the system for practical application in rehabilitation and sports-performance monitoring contexts

ENET'Com Room Booking Web Application

2025

Full-Stack Development

ReactJS, Laravel, PHP, SQL

- Designed and developed a full-stack room reservation platform for ENET'Com student clubs
- Implemented authentication, booking logic, and role-based access control for admin and student users
- Built a relational database schema to manage room availability, reservations, and conflict prevention across clubs

EDUCATION

ENET'Com – National Engineering School of Electronics and Communications <i>B.Eng. in Electronics and Communications Engineering</i>	Sept 2024 – Present <i>Sfax, Tunisia</i>
IPEIM – Preparatory Institute for Engineering Studies <i>Mathematics, Physics & Engineering Sciences</i>	Sept 2021 – Sept 2024 <i>Monastir, Tunisia</i>
Lycée de Teboulba <i>Baccalauréat in Experimental Sciences – High Distinction (Mention Très Bien)</i>	Sept 2021 <i>Monastir, Tunisia</i>

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

General Secretary <i>ENET'Com Junior Entreprise (ECJE)</i> • Oversee administrative and organizational coordination of the junior enterprise, chair meetings, and draft official minutes • Lead internal structuring initiatives across operational poles	June 2025 – Present <i>Sfax, Tunisia</i>
Various Roles – Project Coordination & Marketing <i>ENET'Com Junior Entreprise (ECJE)</i> • Deputy Project Process Manager (Feb–May 2025): coordinated project delivery, tracked progress across active missions, and improved internal processes for team efficiency • Active Member, Marketing & Events (Oct 2024–Feb 2025): created marketing materials and visual content, sourced partnerships, and co-organized events	Oct 2024 – May 2025 <i>Sfax, Tunisia</i>

VOLUNTEER EXPERIENCE

Camp Counsellor (Moniteur) <i>ONET Teboulba – Organisation Nationale de l'Enfance Tunisienne</i> • Led and facilitated workshops (theatre, music, dance, arts & crafts, brain games) for children aged 8–14 during national summer camps • Instilled core values such as self-reliance, teamwork, and civic responsibility through structured group activities	<i>Teboulba, Tunisia</i>
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RESEARCH INTERESTS

Passionate about applying Deep Learning and Edge AI to solve real-time challenges in Biomedical Signal Processing and embedded healthcare systems, with broader interests in IoT and full-stack development for research-driven applications.